Chapter 11: Grandmother's Quilt

Worksheet 11: Grandmother's Quilt

Learning Objective: Explore how shapes (squares, triangles) combine to form patchwork patterns; relate parts of a shape to fractions and calculate area of simple patchwork pieces.

Worked Example:
A quilt square of side 20 cm is divided into 4 equal smaller squares. What fraction of the big square does each small square represent, and what is its area?

Solution: Each small square represents $\frac{1}{4}$ of the big square. Side of small square = 10 cm, area = $10 \times 10 = 100$ sq cm.

Questions (1–20)

1. If a square is cut into 2 equal triangles, what fraction of the square does each triangle represent?

2. A quilt patch is divided into 4 equal triangles by its two diagonals. What fraction is each triangle?

3. Find the area of a square quilt patch with side 8 cm.

4. Find the area of a rectangular quilt patch 12 cm by 5 cm.

5. How many small squares of side 5 cm are needed to fully tile a square patch of side 10 cm?

6. A quilt is made of 9 identical square patches arranged in a 3×3 grid, each of side 15 cm. Find the side length and area of the whole quilt.

7. Grandmother uses $\frac{2}{8}$ of her cloth pieces for red squares and $\frac{3}{8}$ for blue squares. What fraction is used for other colours?

8. A quilt patch is a square of area 144 sq cm. Find the length of its side.

9. A quilt design uses triangles each with area 50 sq cm. If 8 such triangles are used, find the total area covered.

10. A quilt has 6 rows of square patches with 5 patches in each row. How many patches in total?

11. Grandmother's quilt is made of 20 square patches, each of side 10 cm, arranged in a 4×5 grid. Find the total area of the quilt.

12. A quilt patch is divided into 4 equal triangles. 1 is red, 1 is yellow, and 2 are green. What fraction of the patch is green?

13. A quilt strip is made by joining 12 square patches of side 25 cm in a row. Find the total length of the strip.

14. Each square patch needs 4 corner stitches. How many stitches are needed for 15 patches (assuming no shared stitching)?

15. A quilt has a repeating pattern of 3 squares and 2 triangles. If there are 50 shapes total, how many squares and triangles are there?

16. A square quilt patch of side 16 cm is cut into 4 equal smaller squares, and one of those is further cut into 4 equal triangles. What fraction of the original large patch does one such triangle represent?

17. Grandmother has 60 square patches and wants to arrange them into a rectangular quilt with more rows than columns. Give two possible row $\times$ column arrangements.

18. A quilt has alternating red and white square patches like a chessboard, in an 8×8 grid (64 patches). How many red patches and how many white patches are there?

19. Look at a quilt, bedsheet, or cloth pattern at home. Describe the shapes used and whether they repeat in a pattern.

20. Design (describe) a 2×2 grid of 4 square patches using 2 colours in a pattern.

Reflection Question: Old quilts were often made by joining many small leftover cloth pieces. Why do you think this was a useful and creative way to use cloth? Write your thoughts.

Real-Life Application Question: Find any object at home made up of repeated smaller shapes joined together (a tiled floor, a woven basket, a quilt). Describe the shapes and how many you can count.